

Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor

Abstract:

Introduce the project as a business intelligence and financial analytics solution that analyses product profitability, division performance, cost efficiency, and revenue contribution using data analytics and an interactive Streamlit dashboard. Mention that KPIs, Pareto analysis, and machine learning support strategic decision-making.

Introduction:

This project presents a Product Line Profitability and Margin Performance Analysis system that combines data preprocessing, profitability metrics, exploratory data analysis, machine learning, and an interactive Streamlit dashboard. The application enables stakeholders to monitor financial performance, compare divisions, evaluate cost structures, and identify products that contribute significantly to organizational profitability.

The primary objective is to transform raw transactional data into meaningful business intelligence that supports pricing optimization, cost management, inventory planning, and strategic product portfolio decisions.

Dataset Description:

Dataset:

Nassau Candy Distributor Sales Dataset

- Row ID
- Order ID
- Order Date
- Ship Date
- Ship Mode
- Customer ID
- Country/Region
- City
- State/Province
- Postal Code
- Division
- Region
- Product ID
- Product Name
- Sales
- Units
- Cost
- Gross Profit

Data Preprocessing :

- Removal of duplicate records
- Validation of sales and cost values
- Removal of zero-sales and invalid profit records
- Handling missing unit values
- Standardization of product names
- Standardization of division labels
- Conversion of financial columns into numerical format
- Creation of profitability metrics
- Feature engineering for dashboard KPIs

Exploratory Data Analysis (EDA) :

Product Profitability Overview:

Product Margin Leaderboard

Interpretation

The Product Margin Leaderboard ranks products according to Gross Margin Percentage and Total Profit. The objective is to identify products that generate the highest financial returns while maintaining healthy profit margins. This visualization enables stakeholders to recognize top-performing products and compare profitability across the entire product portfolio.

Key Insight

Products appearing consistently at the top of the leaderboard contribute significantly to organizational profitability and should receive continued investment, inventory prioritization, and marketing support. Products with low rankings may require pricing adjustments or cost optimization.

Product-Level Margin Leaderboard

	Product Name	Total_sales	Total_units	Total_cost	Total_profit	Gross_Margin_%	Profit_per_unit	Revenue_Contribution_%	Profit_Contribution_%
1	Everlasting Gelatoque	\$130	18	\$26	\$104	80.0%	\$5.78	0.1%	0.1%
2	Wain Toffee	\$76	17	\$17	\$60	77.6%	\$3.53	0.2%	0.2%
11	Wonka Bar - Nutty Crunch Surprise	\$23,075	8755	\$6,755	\$16,320	71.2%	\$2.45	16.6%	18.0%
13	Wonka Bar - GnomishGumpshinus	\$27,875	7743	\$8,117	\$19,758	69.4%	\$2.56	19.7%	20.7%
9	Wonka Bar - Fudge Walkers	\$24,880	8934	\$8,287	\$16,593	66.7%	\$2.42	17.6%	17.9%
12	Wonka Bar - Triple Doodle Caramel	\$26,480	7706	\$9,875	\$16,605	63.1%	\$2.43	20.1%	19.9%
10	Wonka Bar - Milk Chocolate	\$26,868	8287	\$9,424	\$17,443	64.9%	\$2.11	18.9%	18.7%
5	Luffy Taffy	\$84	27	\$20	\$63	62.9%	\$1.24	0.0%	0.0%
8	Fizzy Spring Drinks	\$76	21	\$23	\$47	60.9%	\$2.25	0.1%	0.1%
14	Wonka Gum	\$988	476	\$187	\$791	52.0%	\$6.65	0.4%	0.8%

Profit Contribution Chart:

Interpretation

The Profit Contribution Chart illustrates the percentage contribution of each product toward overall organizational profit. The visualization helps determine whether profitability is evenly distributed or concentrated among a limited number of products.

Key Insight

Gross Margin % by Product

Product Name	Gross Margin %
Chocolate Chip Cookies	75%
Vanilla Cakes	74%
Buttercream Cupcakes	72%
Chocolate Brownies	71%
Vanilla Cupcakes	70%
Chocolate Brownies	69%
Vanilla Cakes	68%
Buttercream Cupcakes	67%
Chocolate Chip Cookies	66%
Vanilla Cakes	65%
Buttercream Cupcakes	64%
Chocolate Brownies	63%
Vanilla Cupcakes	62%
Chocolate Brownies	61%
Vanilla Cakes	60%
Buttercream Cupcakes	59%
Chocolate Chip Cookies	58%
Vanilla Cakes	57%
Buttercream Cupcakes	56%
Chocolate Brownies	55%
Vanilla Cupcakes	54%
Chocolate Brownies	53%
Vanilla Cakes	52%
Buttercream Cupcakes	51%
Chocolate Chip Cookies	50%
Vanilla Cakes	49%
Buttercream Cupcakes	48%
Chocolate Brownies	47%
Vanilla Cupcakes	46%
Chocolate Brownies	45%
Vanilla Cakes	44%
Buttercream Cupcakes	43%
Chocolate Chip Cookies	42%
Vanilla Cakes	41%
Buttercream Cupcakes	40%
Chocolate Brownies	39%
Vanilla Cupcakes	38%
Chocolate Brownies	37%
Vanilla Cakes	36%
Buttercream Cupcakes	35%
Chocolate Chip Cookies	34%
Vanilla Cakes	33%
Buttercream Cupcakes	32%
Chocolate Brownies	31%
Vanilla Cupcakes	30%
Chocolate Brownies	29%
Vanilla Cakes	28%
Buttercream Cupcakes	27%
Chocolate Chip Cookies	26%
Vanilla Cakes	25%
Buttercream Cupcakes	24%
Chocolate Brownies	23%
Vanilla Cupcakes	22%
Chocolate Brownies	21%
Vanilla Cakes	20%
Buttercream Cupcakes	19%
Chocolate Chip Cookies	18%
Vanilla Cakes	17%
Buttercream Cupcakes	16%
Chocolate Brownies	15%
Vanilla Cupcakes	14%
Chocolate Brownies	13%
Vanilla Cakes	12%
Buttercream Cupcakes	11%
Chocolate Chip Cookies	10%
Vanilla Cakes	9%
Buttercream Cupcakes	8%
Chocolate Brownies	7%
Vanilla Cupcakes	6%
Chocolate Brownies	5%
Vanilla Cakes	4%
Buttercream Cupcakes	3%
Chocolate Chip Cookies	2%
Vanilla Cakes	1%
Buttercream Cupcakes	0%
Chocolate Brownies	-1%
Vanilla Cupcakes	-2%
Chocolate Brownies	-3%
Vanilla Cakes	-4%
Buttercream Cupcakes	-5%
Chocolate Chip Cookies	-6%
Vanilla Cakes	-7%
Buttercream Cupcakes	-8%
Chocolate Brownies	-9%
Vanilla Cupcakes	-10%
Chocolate Brownies	-11%
Vanilla Cakes	-12%
Buttercream Cupcakes	-13%
Chocolate Chip Cookies	-14%
Vanilla Cakes	-15%
Buttercream Cupcakes	-16%
Chocolate Brownies	-17%
Vanilla Cupcakes	-18%
Chocolate Brownies	-19%
Vanilla Cakes	-20%
Buttercream Cupcakes	-21%
Chocolate Chip Cookies	-22%
Vanilla Cakes	-23%
Buttercream Cupcakes	-24%
Chocolate Brownies	-25%
Vanilla Cupcakes	-26%
Chocolate Brownies	-27%
Vanilla Cakes	-28%
Buttercream Cupcakes	-29%
Chocolate Chip Cookies	-30%
Vanilla Cakes	-31%
Buttercream Cupcakes	-32%
Chocolate Brownies	-33%
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Chocolate Brownies	-35%
Vanilla Cakes	-36%
Buttercream Cupcakes	-37%
Chocolate Chip Cookies	-38%
Vanilla Cakes	-39%
Buttercream Cupcakes	-40%
Chocolate Brownies	-41%
Vanilla Cupcakes	-42%
Chocolate Brownies	-43%
Vanilla Cakes	-44%
Buttercream Cupcakes	-45%
Chocolate Chip Cookies	-46%
Vanilla Cakes	-47%
Buttercream Cupcakes	-48%
Chocolate Brownies	-49%
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Chocolate Brownies	-51%
Vanilla Cakes	-52%
Buttercream Cupcakes	-53%
Chocolate Chip Cookies	-54%
Vanilla Cakes	-55%
Buttercream Cupcakes	-56%
Chocolate Brownies	-57%
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Chocolate Brownies	-59%
Vanilla Cakes	-60%
Buttercream Cupcakes	-61%
Chocolate Chip Cookies	-62%
Vanilla Cakes	-63%
Buttercream Cupcakes	-64%
Chocolate Brownies	-65%
Vanilla Cupcakes	-66%
Chocolate Brownies	-67%
Vanilla Cakes	-68%
Buttercream Cupcakes	-69%
Chocolate Chip Cookies	-70%
Vanilla Cakes	-71%
Buttercream Cupcakes	-72%
Chocolate Brownies	-73%
Vanilla Cupcakes	-74%
Chocolate Brownies	-75%
Vanilla Cakes	-76%
Buttercream Cupcakes	-77%
Chocolate Chip Cookies	-78%
Vanilla Cakes	-79%
Buttercream Cupcakes	-80%
Chocolate Brownies	-81%
Vanilla Cupcakes	-82%
Chocolate Brownies	-83%
Vanilla Cakes	-84%
Buttercream Cupcakes	-85%
Chocolate Chip Cookies	-86%
Vanilla Cakes	-87%
Buttercream Cupcakes	-88%
Chocolate Brownies	-89%
Vanilla Cupcakes	-90%
Chocolate Brownies	-91%
Vanilla Cakes	-92%
Buttercream Cupcakes	-93%
Chocolate Chip Cookies	-94%
Vanilla Cakes	-95%
Buttercream Cupcakes	-96%
Chocolate Brownies	-97%
Vanilla Cupcakes	-98%
Chocolate Brownies	-99%
Vanilla Cakes	-100%

Profit Contribution by Product

Product Name	Profit Contribution %
Chocolate Chip Cookies	35.0%
Vanilla Cakes	30.0%
Buttercream Cupcakes	25.0%
Chocolate Brownies	20.0%
Vanilla Cupcakes	15.0%
Chocolate Brownies	10.0%
Vanilla Cakes	5.0%
Buttercream Cupcakes	0.0%
Chocolate Chip Cookies	-5.0%
Vanilla Cakes	-10.0%
Buttercream Cupcakes	-15.0%
Chocolate Brownies	-20.0%
Vanilla Cupcakes	-25



Interpretation

Key Insight

A bar chart titled "Revenue vs Profit Share by Division" comparing the revenue and profit shares of three divisions: Chocolate, Other, and Sugar. The Y-axis represents the percentage share, ranging from 0 to 100. The X-axis lists the divisions. For each division, there are two bars: a dark blue bar for "Revenue Share %" and a light blue bar for "Profit Share %".

Division	Revenue Share %	Profit Share %
Chocolate	92	95
Other	6	4
Sugar	0.5	0.5

Interpretation

Key Insight

Divisions with higher average margins demonstrate effective pricing strategies and cost management. Lower-margin divisions may require operational review, supplier negotiations, or revised pricing policies.



Cost vs Margin Diagnostics

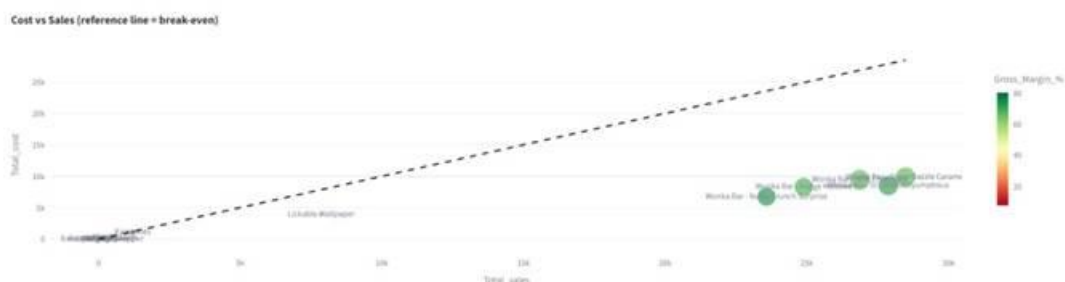
Cost-sales scatter plots

Interpretation

The scatter plot compares product cost against sales value to evaluate pricing efficiency and identify products with disproportionately high costs.

Key Insight

Products located in high-cost and low-sales regions indicate potential financial risks and should be reviewed for procurement optimization, supplier negotiation, or discontinuation.



Margin risk flags

Interpretation

Margin Risk Flags classify products according to their financial health using predefined Gross Margin thresholds. Products with exceptionally low margins are automatically highlighted for management attention.

Key Insight

Risk indicators assist decision-makers in identifying financially vulnerable products requiring immediate pricing review, cost reduction, or strategic repositioning.

Cost-heavy / margin-poor		Needs repricing (>15% margin)			Needs cost renegotiation (>60% CTS)		
1		1			1		
	Product Name	Total_sales	Total_cost	Total_units	Total_profit	Gross_Margin_%	Cost_to_Sales_%
4	Kacookies	1205.75	1113	371	92.75	7.6923	92.3077
2	Fun Dip	12	7.2	8	4.8	40	60
3	Nerdz	15	8	10	7	46.6667	53.3333
6	Sweetarts	91.5	32.8	41	26.7	46.6667	53.3333
9	Lickable Waflpaper	7960	3930	393	3930	50	50
14	Wonka Gum	597.5	286.8	419	310.7	52	48
1	Fizzy Lifting Drinks	75.75	31.5	21	47.25	60	40
5	Laffy Taffy	53.73	20.25	27	33.48	62.3136	37.6864
10	Wonka Bar - Milk Chocolate	26867.75	9424.38	8267	17443.37	64.9231	35.0769
12	Wonka Bar - Triple Dazzle Caramel	28485	9814.8	7596	18670.2	65.3333	34.6667

Profit Concentration Analysis

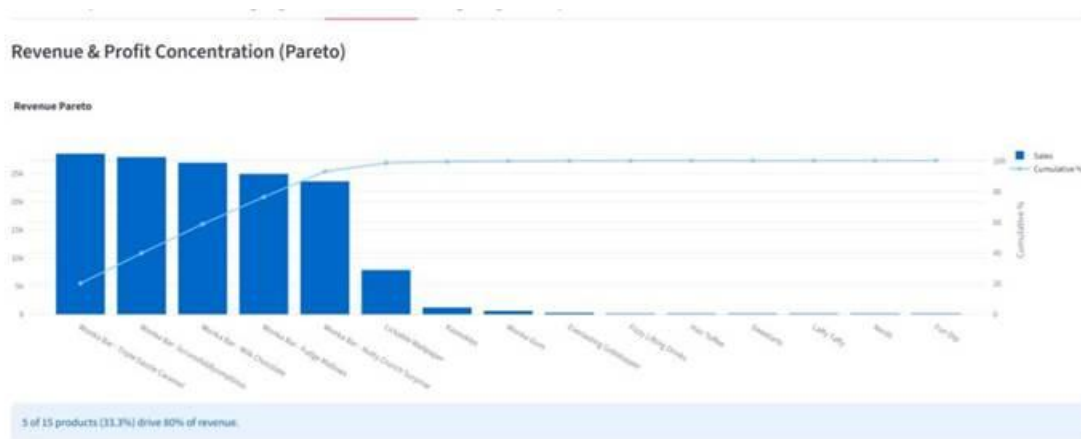
Pareto Analysis

Interpretation

Pareto Analysis applies the 80/20 principle to determine the proportion of products responsible for the majority of revenue and profit. The cumulative contribution curve highlights products with the greatest financial impact.

Key Insight

A relatively small percentage of products typically generates the majority of organizational profit. Monitoring these products is essential for minimizing dependency risks while maximizing business performance.



Dependency Indicators

Interpretation :

Dependency Indicators evaluate the organization's reliance on a limited number of high-performing products by measuring cumulative revenue and profit concentration.

Key Insight

High dependency increases operational risk if demand declines or supply disruptions occur. Diversifying profitability across a broader product portfolio improves long-term business resilience.

State/Province Dependency Risk

Top location: California

19.7% of revenue

Low dependency risk

	State/Province	Total_sales	Total_units	Total_profit	Revenue_%	Units_%
1	California	27917.4	7567	7567	18479.42	19.6901
39	New York	15541.03	4234	4234	10222.44	10.9613
51	Texas	13416.09	3724	3724	8909.53	9.4624
43	Pennsylvania	8027.03	2153	2153	5225.47	5.6825
55	Washington	6921.15	1883	1883	4566.64	4.8815
11	Illinois	6896.96	1845	1845	4517.68	4.8658
29	Ohio	6768.95	1779	1779	4413.03	4.7741
10	Florida	4804.02	1379	1379	3287.11	3.3883
2	Arizona	3587.55	862	862	2290.11	2.5303
35	North Carolina	3450.86	963	963	2331.16	2.5431

Machine Learning Insights

The project incorporates Machine Learning techniques to enhance traditional profitability analysis by automatically identifying patterns among products based on their financial performance. K-Means Clustering is used to segment products into groups with similar profitability characteristics, enabling more effective business decision-making and product portfolio management.

Product Segmentation using K-Means Clustering

Interpretation

This visualization presents the results of K-Means clustering, where products are grouped according to their profitability characteristics using features such as Gross Margin Percentage, Total Sales, Profit per Unit, and Revenue Contribution. Each cluster represents products with similar financial behaviour, allowing stakeholders to identify high-performing products, growth opportunities, and underperforming product categories.

Key Insight

The clustering results reveal distinct profitability segments within the product portfolio. High-value clusters contain products generating strong revenue and profit, while lower performing clusters indicate products requiring pricing adjustments, cost optimization, or strategic review. This segmentation supports targeted business strategies instead of applying uniform decisions across all products.



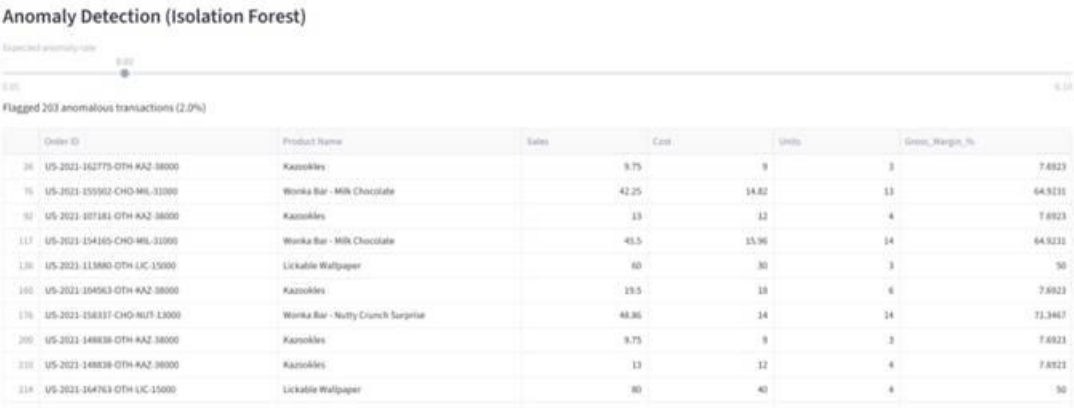
Anomaly Detection (Isolation Forest)

Interpretation

This visualization highlights products classified as either **Normal** or **Anomalous** based on their financial characteristics. The Isolation Forest algorithm evaluates multiple profitability-related features, including sales, cost, gross margin percentage, profit per unit, and revenue contribution, to detect observations that deviate significantly from normal business patterns. Anomalous products are visually distinguished to help stakeholders quickly identify records requiring further analysis.

Key Insight

The detected anomalies represent products exhibiting unusual financial behavior compared to the rest of the portfolio. These products may indicate exceptional business opportunities, pricing inconsistencies, operational inefficiencies, data quality issues, or unexpected market conditions. Early identification of such products enables organizations to take corrective actions and improve financial decision-making.



Sales Forecast (Linear Trend, 3 months)

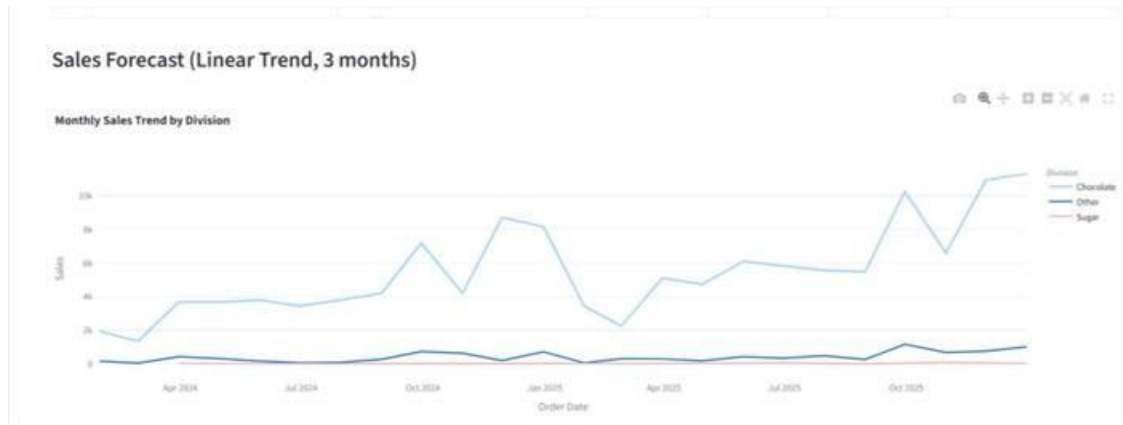
Three-Month Sales Forecast

Interpretation

This visualization presents the historical sales trend alongside the projected sales values for the upcoming three months using a Linear Trend forecasting model. The trend line is generated by fitting a linear relationship to historical sales data and extending it into the forecast period. The objective is to estimate future sales performance based on past business trends while providing an easily interpretable forecasting approach.

Key Insight

The forecast indicates the expected direction of future sales, allowing stakeholders to anticipate potential growth or decline in business performance. An upward trend suggests increasing market demand and revenue opportunities, whereas a downward trend may indicate the need for strategic interventions such as promotional campaigns, pricing adjustments, or inventory optimization.



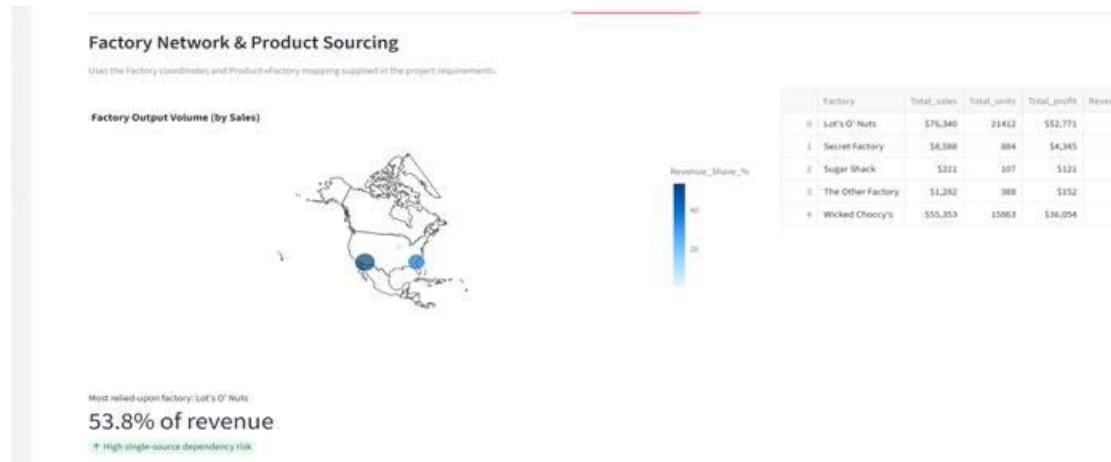
Factory Network & Logistics Analysis

Interpretation

This visualization presents the geographic distribution of manufacturing facilities and their associated product sourcing locations. By integrating factory coordinates with product-level financial data, the dashboard illustrates how products are allocated across different factories and provides an overview of the manufacturing network. The analysis helps stakeholders understand factory utilization, production coverage, and sourcing relationships while supporting logistics planning and supply chain visibility.

Key Insight

The Factory Network & Product Sourcing analysis highlights the contribution of each manufacturing facility to the overall production network and reveals how products are distributed across factories. This insight supports better production planning, resource allocation, supply chain optimization, and operational decision-making by improving visibility into manufacturing operations and reducing dependency on individual production facilities.



Factory-to-Destination Flow Analysis

Interpretation

This visualization represents the indicative movement of products from manufacturing facilities to their respective destination regions. Lines connecting factories and destinations illustrate the distribution pathways within the supply chain, providing an overview of how products are delivered across different markets. The objective is to understand the geographical relationship between production facilities and customer locations while identifying major distribution routes.

Key Insight

The Factory → Destination Flow highlights the connectivity between manufacturing facilities and customer regions, enabling stakeholders to identify primary distribution corridors and evaluate the overall structure of the supply network. Understanding these logistics pathways supports improved transportation planning, efficient product allocation, and enhanced operational coordination.—

Indicative Factory → Destination Flow

Top 25 Factory → State Sales Flows (line width not to scale)



Note: destination points are approximate state/province centroids, not exact delivery addresses, so distances are indicative only — useful for spotting which factories serve distant regions, not for precise route optimisation.

	Factory	Avg_Distance_km	Total_sales	Revenue_Share_%
5	Lot's O' Nuts	2,307	\$76,340	53.8%
1	Secret Factory	1,309	\$8,588	6.1%
2	Sugar Shack	1,791	\$221	0.2%
3	The Other Factory	1,338	\$1,282	0.9%
6	Wicked Choccy's	1,747	\$55,353	39.0%

Data quality note: the raw file's Ship Date values are 3-4 years after the matching Order Date on every row, which is not a plausible real shipment lag. Treat this as a placeholder field rather than a real fulfillment-time metric until it is corrected at the source.

Key Performance Indicators

- **Gross Margin (%) :**

Interpretation: Measures profitability relative to sales revenue and evaluates pricing efficiency.

- **Profit per Unit :**

Interpretation: Indicates the average profit generated by each unit sold.

- **Revenue Contribution :**

Interpretation: Measures the percentage contribution of each product toward total organizational revenue.

- **Profit Contribution :**

Interpretation: Measures the percentage contribution of each product toward total organizational profit.

Streamlit Dashboard Development :

The Streamlit web application was developed to provide an interactive business intelligence platform for profitability analysis. The dashboard integrates financial KPIs, product rankings, division-level analytics, cost diagnostics, and Pareto analysis into a unified interface. The application includes:

- Date range selector
- Division filter
- Margin threshold slider
- Product search

Filters

Order Date range

2024/01/02 – 2025/12/31

Division

Chocolate × Other × Sugar ×

Minimum Gross Margin %

-50

Product search

Findings

Summarize the key observations from your analysis, for example:

- Profitability is concentrated among a relatively small set of products.
- High revenue does not always correspond to high profitability.
- Margin performance varies significantly across divisions.
- Cost diagnostics identify products with pricing or procurement inefficiencies.

Interactive dashboards improve the speed and accuracy of financial analysis.

Recommendations

- Prioritize investment in consistently high-margin products.
- Review pricing strategies for products with high sales but low margins.
- Negotiate procurement costs for cost-heavy products.
- Monitor divisions with low financial efficiency.
- Use Pareto insights to diversify the product portfolio and reduce dependency risk.

Conclusion

Conclude by explaining that the project successfully integrates data cleaning, profitability metrics, exploratory data analysis, machine learning (if implemented), and an interactive Streamlit dashboard to transform raw sales data into actionable financial insights. Emphasize that the system supports informed decision-making through dynamic visualizations, KPI monitoring, and strategic business recommendations.